Secure Data Analytics with GenAI and LLMs

Day 1 Participant Workbook

Theme: From AI Literacy to Responsible Analytics
Session focus: Foundations, prompting, literature review, writing, and coding support
Date: 16 July 2026
Duration: 3 hours
Audience: Faculty, graduate students, researchers, analysts, academic staff, professionals, and curious beginners
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> ***AI can accelerate parts of your work, but it does not remove your responsibility to think, verify, protect data, cite sources, and make the final decision.***

How to Use This Workbook

This is not a book to read passively. During the workshop, you will ***predict, discuss, prompt, critique, revise, verify, and reflect***. Write directly in the spaces provided. If you are new to AI, complete the **Foundation Track**. If you already use AI, add the ***Challenge Track*** to test whether your current practice is genuinely reliable.

The learning cycle used throughout the session is:

Stage	What you will do
Explain	Build a clear mental model in ordinary language.
Demonstrate	Observe a worked example or live AI interaction.
Practice	Apply the idea to a realistic research, writing, or coding task.
Evaluate	Inspect the output rather than accepting it automatically.
Reflect	Decide what you will keep, change, or avoid in your own work.

Learning Outcomes

By the end of this session, you should be able to:

1. explain *AI*, *machine learning*, *deep learning*, *generative AI*, *foundation models*, and *large language models* in plain language;
2. describe how an LLM generates text and why a fluent answer may still be wrong;
3. use the *TASK framework* to write clearer prompts for research and analysis;
4. use the *TRACE check* to evaluate AI-generated content;
5. apply a *Human–AI–Human workflow* to literature review, academic writing, and coding support; and
6. protect sensitive information, document important decisions, and disclose AI use when required.

Session Map

Time	Segment	Main activity
0:00–0:10	Welcome and diagnostic	Poll, self-rating, pair-share
0:10–0:45	Module 1A: AI and GenAI foundations	Concept sorting and myth check
0:45–1:05	Module 1B: How LLMs work and fail	Human token-prediction demo and verification exercise
1:05–1:15	Break	One-minute reflection
1:15–1:55	Module 2: Better prompts	Prompt makeover and paired testing
1:55–2:35	Module 3A: Literature review and writing	Evidence workflow and claim–evidence matrix
2:35–2:55	Module 3B: Coding support	Explain–debug–test mini-lab
2:55–3:00	Exit ticket	Personal action commitment

Before we begin let us know your starting point

Activity 0.1 — Quick Diagnostic

Rate yourself from **1 = not yet confident** to **5 = very confident**.

Statement	Before	After
I can explain the difference between AI and GenAI.		
I understand, at a useful level, how an LLM generates an answer.		
I can write a prompt that specifies context, evidence, and output requirements.		
I can identify when an AI output requires verification.		
I can use AI responsibly in literature review and academic writing.		
I can use AI to explain, debug, or test code without trusting it blindly.		

My most important question for today: _____
_____

One AI task I already perform, or would like to perform: _____
_____

Activity 0.2 — Think–Pair–Share

In one minute, complete this sentence:

> *"AI is useful when _____, but risky when _____
> _____."*

Compare your answer with a partner. Do you emphasise the same benefit and risk?

Module 1 — Foundational AI and GenAI Literacy

1.1 A Practical Family of Terms

Artificial intelligence (AI) is the broad field concerned with computer systems that perform tasks associated with intelligent behaviour, including prediction, classification, recognition, generation, planning, and control. Or we can also say -

“AI is software that performs tasks which, if a human did them, we would call intelligent — recognising a face, answering a question, writing a paragraph, predicting a trend”

Generative AI is the part of AI primarily designed to generate new content such as text, images, audio, video, code, or synthetic data.

A **large language model (LLM)** is a language-focused model trained on very large amounts of text and related data;

A **foundation model** is a broad-capability model that can be adapted to many downstream tasks. These categories overlap, so use them as a practical map rather than perfectly sealed boxes.[3]

Term	Plain-language meaning	Familiar example	Important distinction
AI	The broad umbrella of machines performing prediction, recognition, generation, or decision-support tasks.	Fraud detection, route planning, speech recognition	Not all AI generates content.
Machine learning (ML)	Systems learn useful patterns from data rather than relying only on hand-written rules.	Email spam classification	ML is one way to build AI.
Deep learning	ML using layered neural networks to learn complex patterns.	Image recognition, speech-to-text	Deep learning is a subset of ML.
Generative AI (GenAI)	AI that produces new content.	Text, images, code, music, summaries	Generated does not mean verified.
Foundation model	A broad model that can be adapted to many tasks.	A model supporting chat, extraction, coding, and classification	“Foundation model” refers to breadth and adaptability.

LLM	A foundation model specialised in understanding and generating language.	A conversational assistant that drafts or explains text	It generates likely language; it is not a database of guaranteed facts.
AI tool or application	A user-facing product built around one or more models plus interfaces, search, files, or other functions.	A research assistant or coding assistant	The tool is more than the underlying model.

Activity 1.1 — Concept Sort

For each example, identify the most specific useful category: **AI, ML, GenAI, or LLM application**. Some examples fit more than one; be ready to defend your choice.

Example	Your category	Reason
A bank system predicts whether a transaction is fraudulent.		
A tool creates an illustration from a written description.		
A chatbot summarises a supplied journal article.		
A thermostat learns the household's preferred schedule.		
A model drafts Python code from a natural-language instruction.		

> ***Challenge Track:*** *For one example, separate the* ***model****, the* ***application****, the* ***data****, and the* ***human decision****. Why does the distinction matter when something goes wrong?*

1.2 What an LLM is Doing

A useful introductory mental model is that an LLM receives a prompt, converts the text into smaller units called **tokens**, represents those tokens numerically, estimates probable continuations from patterns learned during training, selects a continuation, and repeats the process. This next-token account is not the whole technical story, but it explains why one model can complete text, answer questions, translate, summarise, and generate code.[11]

Step	Simplified process	Practical implication
1. **Input**	The model receives your instructions and available context.	Missing or ambiguous context can change the answer.
2. **Tokenisation**	Text is divided into tokens, which may be words or word-parts.	Length limits are measured in tokens, not simply pages.
3. **Pattern-based prediction**	The model estimates likely next tokens using learned patterns and current context.	The answer can sound natural without being factually grounded.
4. **Generation**	It chooses a token and continues, one step at a time.	Different runs may produce different valid or invalid outputs.
5. **Completion**	The tool presents the generated response, sometimes with search or other tool results.	You must know whether the answer came from supplied evidence, web search, a tool, or model memory.

Activity 1.2 — Become the Language Model

Complete each sequence individually, then compare answers with the class.

7 "The capital of Ghana is _____."
8 "Twinkle, twinkle, little _____."
9 "The research findings were statistically significant; however, _____."
10 "To improve the reliability of this analysis, the researcher should _____."

Debrief:

- Which completion had one obvious answer?
- Which allowed several reasonable answers?
- What prior patterns did you use?
- Could the most natural completion still be false in a specific context?

> ***Key lesson:*** *A language model is optimised to produce a plausible continuation. Plausibility and factual correctness can overlap, but they are not the same thing.*

Go Further — Randomness, Temperature, and Repeatability

Many systems can vary how conservatively or creatively they select among likely tokens. This is often described using a **temperature** setting. Lower settings generally favour more probable continuations; higher settings permit more variation. Even with careful settings, model outputs may not be perfectly repeatable.[11] For consequential tasks, preserve the model or tool name, date, relevant settings, prompt, supplied sources, and final human decision.

1.3 Capabilities: Use the Right Verb

Avoid saying that AI simply "knows" or "does research." Name the actual operation.

Useful operation	Example	What remains human work
Generate	Produce possible research questions.	Select a worthwhile, feasible, ethical question.
Transform	Rephrase a paragraph for clarity.	Preserve meaning, voice, evidence, and appropriate attribution.
Summarise	Condense a supplied report.	Check coverage, omissions, nuance, and accuracy.
Extract	Pull study characteristics into a table.	Validate every field against the source.
Classify	Apply proposed screening labels.	Define criteria, audit errors, and resolve uncertainty.
Compare	Contrast methods or arguments.	Decide whether the comparison is fair and evidence-based.
Explain	Explain code or a statistical idea at beginner level.	Confirm the explanation and practise the underlying skill.
Critique	Identify possible weaknesses in an argument.	Judge whether each criticism is valid and important.
Draft	Propose an outline, email, code block, or paragraph.	Own the ideas, revise the content, verify facts, and approve the final form.

1.4 Limits and Risks

UNESCO recommends a human-centred approach to GenAI in education and research, with attention to privacy, ethical validation, pedagogical design, equity, and meaningful human use.[1] NIST's Generative AI Profile likewise frames trustworthy use as a risk-management problem spanning design, use, and evaluation rather than a one-time prompt-writing problem.[2]

Risk	What it may look like	A practical control
Hallucination or confabulation	A plausible but false fact, quotation, reference, result, or explanation	Verify important claims in authoritative or primary sources.
Bias and unfairness	Stereotyped assumptions, missing perspectives, or unequal performance	Ask whose data or viewpoint may be absent; review effects on affected groups.
Privacy loss	Sensitive text, records, code, or identifiers entered into an unsuitable service	Apply the traffic-light data rule before prompting.
Outdated or incomplete knowledge	Missing recent work, local context, or pay walled evidence	State the required date range and use current databases or approved search tools.
False authority	Confident wording encourages acceptance without checking	Require uncertainty, evidence boundaries, and a TRACE review.
Overreliance and skill erosion	Users outsource reading, writing, analysis, or coding they need to understand	Use AI after attempting the task or require explanations and reflection.
Security and code defects	Vulnerable code, unsafe commands, weak validation, or invented libraries	Run tests, inspect dependencies, review security, and never execute unknown code blindly.
Integrity and authorship problems	Undisclosed AI-written content or fabricated citations	Follow institutional and publisher policy; disclose material use where required.

OpenAI describes hallucinations as **plausible but false statements** and notes that systems can be rewarded for guessing rather than admitting uncertainty.[12] Therefore:

! Fluency is not evidence. ***! Confidence is not certainty.***
! A citation-shaped sentence is not proof that the cited source exists or supports the claim.

Activity 1.3 — Trust, Verify, or Do Not Use?

Mark each proposed use as **T = generally low-risk to trust after a quick check**, **V = verify carefully**, or **D = do not use in this way**.

Context and institutional policy may change your answer.

Proposed use	T / V / D	Why?
Brainstorming ten possible workshop titles with no confidential information		
Asking for exact quotations and page numbers from a paper that was not provided		
Pasting identifiable patient records into a public chatbot for summarisation		
Asking an AI assistant to explain an error message, then testing the proposed fix		
Using an AI-generated reference list without opening the sources		
Asking for three alternative structures for a paragraph you wrote		

1.5 The Traffic-Light Data Rule

Before entering any information into an AI tool, classify it.

Level	Typical data	Action
Green — Public or non-sensitive	Published papers, public documentation, invented examples, public code with a compatible licence	May be used in an approved tool, with normal accuracy checks.
Amber — Uncertain, restricted, or internally sensitive	Unpublished drafts, de-identified research data, internal reports, student work, grant ideas	Pause. Check consent, contracts, tool settings, retention, institutional policy, and supervisor or data-owner approval.
Red — Confidential or high-risk	Personal identifiers, patient or participant records, passwords, private keys, confidential peer review, proprietary code, export-controlled or security-sensitive data	Do not enter into an external AI service unless explicitly authorised with appropriate technical and legal protections.

University research guidance highlights confidentiality, intellectual property, unpublished work, plagiarism, equity, and transparent disclosure as central questions for responsible use. [10]

My personal red-line data are: _____

_____

_____

_____

1.6 TRACE: A Five-Step Output Check

Use **TRACE** before relying on an AI output.

Letter	Check	Questions to ask
T — Test the claims	Accuracy and logic	Which statements can be checked? Do calculations, examples, or counterexamples reveal an error?
R — Review the sources	Existence and support	Do the sources exist? Are they primary or authoritative? Does each source actually support the claim?
A — Assess the fit	Relevance and context	Does the answer address my population, jurisdiction, discipline, data, date range, and purpose?
C — Check privacy, bias, and consequences	Safety and fairness	What sensitive data were used? Who may be excluded or harmed? What happens if the output is wrong?
E — Explain the final judgment	Human accountability	What did I accept, revise, or reject, and why? Can I defend the final decision?

One-Minute Break Reflection

The most important difference between **plausible** and **reliable** is: _____

_____

_____

Module 2 — Better Prompts for Research and Analysis

2.1 What a Prompt Really Is

A prompt is not a magic spell. It is a **task specification** supplied to a probabilistic system. Effective prompting makes the intended work, evidence, boundaries, and output easier to understand. Good practice begins with a clear instruction, adds relevant context, specifies the desired format, uses examples where useful, divides complex work into stages, and improves the prompt through evaluation.[4] [5]

» A longer prompt is not automatically better.
» A better prompt contains **useful information**.

2.2 The TASK Prompt Framework

Element	Meaning	Questions
T — Task	State the operation and decision the output should support.	What exactly should the AI do? What should it not do?
A — Audience and aim	Identify who will use the result and why.	Beginner or specialist? Teaching, publication, policy, or exploration?
S — Sources and context	Supply evidence, definitions, data, assumptions, and boundaries.	Which documents or dataset may it use? What is unknown?
K — Key output requirements	Specify structure, length, tone, criteria, uncertainty, and quality checks.	Table or prose? Citation rules? Date range? What must be flagged?

Weak Prompt

> *Tell me about AI in education.*

The task is broad, the audience is unknown, the evidence boundary is missing, and the output cannot be evaluated against clear criteria.

Improved TASK Prompt

> 1. ***Task:*** *Using only the two policy documents pasted below, compare their recommendations on generative AI in higher education.*
> 2. ***Audience and aim:*** *Write for university lecturers who must design a one-page departmental discussion brief.*
> 3. ***Sources and context:*** *Treat the supplied documents as the full evidence set. Do not add facts from memory. If a requested point is not supported, write "not stated in the supplied sources."*
> 4. ***Key output requirements:*** *Produce a table with rows for privacy, assessment, equity, staff development, and disclosure. For each row, state the position of Document A, the position of Document B, one similarity, one difference, and the exact page or section supporting the comparison. End with three unresolved questions. Keep the brief under 700 words.*

This prompt is stronger because its output can be inspected. It does not guarantee correctness; it makes errors easier to detect.

2.3 Prompting as an Iterative Research Conversation

Use a four-stage loop:

Stage	Action	Example question
1. Frame	Define the task and evidence boundary.	"What decision will this analysis support?"
2. Generate	Ask for an initial output.	"Create a comparison table using only the supplied text."
3. Evaluate	Apply criteria and TRACE.	"Which claims lack direct support?"
4. Refine	Correct the prompt or supply missing evidence.	"Revise only rows 2 and 4 and quote the supporting sentences."

Do not ask the model to "improve" an answer without defining improvement.

Name the criterion: **accuracy, relevance, completeness, clarity, methodological fit, fairness, concision, reproducibility, or source support**.

Activity 2.1 — Prompt Makeover

Rewrite this prompt using TASK:

> ***"Give me a literature review about social media and students."***

TASK element	Your improved content
Task	
Audience and aim	
Sources and context	
Key output requirements	

Foundation Track: Specify a population, outcome, date range, and output table.

Challenge Track: Also define inclusion and exclusion criteria, evidence limitations, uncertainty language, and what the model must do when evidence is missing.

Pair Test

Exchange prompts with a partner. Without asking the author, underline any term that permits multiple interpretations. Then answer:

1. What evidence may the AI use?
2. What would count as a good answer?
3. How would you detect an invented source?
4. What important constraint is missing?

2.4 High-Value Prompt Patterns for Research and Analysis

Pattern A — Clarify the Research Question

> *I am investigating **[topic]** among **[population/context]**. My intended decision or contribution is **[aim]**. Ask me up to five clarification questions about scope, concepts, outcomes, feasibility, and evidence before proposing three researchable questions. Do not invent literature. For each proposed question, explain what evidence would be needed and one limitation.*

Pattern B — Source-Bounded Summary

> *Summarise the supplied document for **[audience]**. Use only the supplied text. Separate the output into: purpose, method or approach, main claims, supporting evidence, limitations stated by the authors, and unanswered questions. Attach a page or section reference to each substantive point. If information is absent, write "not stated." Do not create citations.*

Pattern C — Compare Arguments

> *Compare Documents A and B on **[issue]**. Distinguish direct statements from your inference. Use columns for claim, Document A evidence, Document B evidence, agreement, disagreement, and confidence. Quote no more than **[number]** words per source. Flag comparisons that are not supported by equivalent evidence.*

Pattern D — Critique an Analysis Plan

> *Act as a critical methods reviewer. Evaluate the analysis plan below against these criteria: alignment with the research question, variable definitions, assumptions, missing-data handling, bias, robustness checks, reproducibility, and interpretation limits. Do not calculate results or assume facts not stated. Produce: (1) strengths, (2) high-priority risks, (3) questions for the analyst, and (4) suggested revisions ranked by impact.*

Pattern E — Analyse Supplied Data Safely

> *I will provide **synthetic or approved non-sensitive data** and a data dictionary. First restate the analytical goal, variables, units, missing-value codes, and assumptions. Then propose an analysis plan. Do not execute or report results until I confirm the plan. When analysing, show validation checks, distinguish descriptive from inferential claims, report uncertainty, and identify findings that require domain review.*

Pattern F — Seek Disconfirming Evidence

> *Review the argument below. Identify the three strongest assumptions, plausible counterarguments, evidence that would change the conclusion, and groups or contexts for which the conclusion may not hold. Do not manufacture sources. Label each point as "supported by supplied evidence," "inference," or "requires external verification."*

2.5 A Prompt Quality Rubric

Score each category from **0 to 2**.

Criterion	0	1	2
Task clarity	Vague	General operation stated	Specific operation and decision stated
Audience and aim	Missing	One is stated	Both are stated and relevant
Evidence boundary	Missing	Some context supplied	Sources, limits, and unknowns are explicit
Output requirements	"Write something"	Basic format given	Structure, criteria, length, and uncertainty rules given
Evaluability	No way to judge	Some criteria implied	Success and verification checks are explicit
Safety	Sensitive data or unsafe action	Risk uncertain	Data classification and safeguards are explicit

Interpretation:

Score	Meaning
0–4	The prompt is likely to produce an answer that is hard to use or evaluate.
5–8	The prompt is workable but needs clearer evidence or quality criteria.
9–12	The prompt is well specified; the output still requires TRACE verification.

Activity 2.2 — Response Critique

Choose one AI response generated during the session. Apply TRACE.

TRACE step	Evidence from the response	Your judgment or action
Test the claims		
Review the sources		
Assess the fit		

Check privacy, bias, and consequences		
Explain the final judgment		

2.6 Prompting Mistakes to Avoid

Mistake	Why it fails	Better practice
"Act as an expert" with no task criteria	A role label cannot replace evidence, scope, or standards.	Specify the task, audience, sources, criteria, and uncertainty.
Asking several complex tasks at once	Errors become difficult to locate.	Decompose: plan, inspect, draft, evaluate, revise.
Asking for citations from memory	The model may invent or misstate sources.	Search authoritative databases; supply sources; verify every reference.
Treating the first answer as final	The initial response may reflect ambiguity or missing context.	Evaluate and refine using explicit criteria.
Pasting all available data	More context can introduce privacy risk and irrelevant noise.	Supply the minimum approved data needed for the task.
Hiding uncertainty	The model may guess to satisfy the request.	Permit "insufficient evidence," request confidence reasons, and ask clarifying questions.
Requesting invisible reasoning as proof	A long explanation can still be wrong.	Ask for concise assumptions, evidence, calculations, tests, and checkable outputs.

> ***Practical rule:*** *Ask for* ***work products you can inspect****—tables, source locations, assumptions, tests, alternative explanations, and uncertainty—not merely confident prose.*

Module 3 — AI for Literature Review, Writing, and Coding Support

3.1 The Human–AI–Human Workflow

Use AI inside a human-controlled process.

Phase	Human responsibility	Appropriate AI support
Human: Frame	Define the question, purpose, standards, data permissions, and expected contribution.	Ask clarifying questions or propose alternatives.
AI: Assist	Provide approved context and supervise bounded tasks.	Discover candidates, transform text, extract fields, suggest structure, explain code, generate tests.
Human: Verify and own	Read sources, validate outputs, revise, cite, disclose, decide, and submit.	Help check consistency after the human review.

The workflow prevents two common errors: beginning with an undefined task and ending with an unverified output.

3.2 AI-Assisted Literature Review

A literature review is not a long AI-generated paragraph. It is a transparent process for **finding, selecting, appraising, comparing, and synthesizing evidence**.

A Defensible Workflow

Stage	Human-led question	Possible AI assistance	Required check
1. Frame	What question, population, concept, context, and date range matter?	Generate candidate terms or frameworks.	Confirm scope with disciplinary knowledge.
2. Plan search	Which databases, keywords, subject headings, and grey-literature sources are appropriate?	Suggest synonyms and Boolean combinations.	Test queries in real databases; document changes.

3. Discover	What papers and authors may be relevant?	Semantic search or citation-network exploration.	Confirm bibliographic details and avoid one-tool coverage claims.
4. Screen	Which records meet pre-defined criteria?	Prioritise records or explain proposed labels.	Humans audit exclusions, ambiguous cases, and error rates.
5. Retrieve and read	What does each full text actually report?	Summarise a supplied paper or locate passages.	Read the paper; check tables, methods, and context.
6. Extract	Which fields are needed consistently?	Populate a structured evidence table.	Compare every extracted field with the source.
7. Appraise	How trustworthy and applicable is each study?	Apply a supplied checklist provisionally.	Human methodological judgment is essential.
8. Synthesise	Where do findings agree, differ, and remain uncertain?	Group themes or compare evidence.	Preserve study differences; avoid vote counting by prose.
9. Cite and report	Can another person trace the review decisions?	Format a draft table or reporting checklist.	Use a reference manager; verify all citations and disclose use where required.

No single AI tool completes a trustworthy literature review. Elicit's official materials describe semantic search, evidence-linked reports, screening, and extraction support.[7] ResearchRabbit describes discovery through citation and author networks and visual maps.[8] These can assist discovery, but **connection, popularity, and machine relevance are not the same as study quality**.

Tool Selection by Job

The examples below are illustrative, not endorsements. Features, access, privacy terms, and pricing change; confirm them before use.

Job	Tool category	Examples	Main caution
Bibliographic database search	Discipline or citation database	PubMed, Scopus, Web of Science, IEEE Xplore, ERIC	Search coverage differs by discipline and subscription.

Broad scholarly discovery	Search engine or open index	Google Scholar, Semantic Scholar, OpenAlex interfaces	Ranking is not a quality appraisal.
Semantic paper discovery and extraction	AI research assistant	Elicit and similar tools	Check coverage and every source-linked claim.
Citation-network exploration	Mapping tool	ResearchRabbit, Connected Papers	A network expands discovery; it does not establish validity.
Reference management	Citation manager	Zotero, EndNote, Mendeley	Imported metadata and styles still need checking.
Full-text analysis	Approved AI tool using supplied documents	Institutionally approved assistant	Respect copyright, confidentiality, and context limits.

Activity 3.1 — Build a Search Concept Table

Case: How does the use of generative AI affect learning outcomes among university students?

Concept	Core term	Synonyms or related terms	Exclusions or boundary notes
Population	University students		
Intervention/exposure	Generative AI use		
Outcome	Learning outcomes		
Context	Higher education		

Now ask an AI assistant to propose additional terms **without claiming that the list is complete**. Check each term against disciplinary language and a real database thesaurus or index.

Literature-Search Prompt

> *My review question is: **[question]**. Build a search-concept table with columns for concept, free-text synonyms, spelling variants, broader terms, narrower terms, and possible subject headings. Do not claim that any subject heading is valid until checked in the named database. Then draft separate Boolean query candidates for **[Database A]** and **[Database***

> ***B]****, explaining which syntax must be verified. Identify concepts that may make the search too narrow.*

Claim–Evidence Matrix

Use this table to prevent "summary drift," where polished synthesis becomes detached from the original studies.

Synthesis claim	Study/source	Exact supporting location	Study context and method	Limitation or conflicting evidence	Verification status

Activity 3.2 — Citation Reality Check

When an AI gives you a reference, verify:

Check	Result
Does the title exist in a scholarly database or publisher site?	
Do author names, year, journal, volume, pages, and DOI match?	
Is the source type appropriate for the claim?	
Does the full text actually support the stated claim?	
Is a newer correction, retraction, or conflicting study relevant?	

Never cite an AI assistant as if it were the original evidence. Locate and cite the source itself, subject to the applicable style and policy.

3.3 AI-Assisted Academic Writing

AI can help at several writing stages, but the scholarly contribution must remain yours. Many institutions and publishers do not treat AI systems as authors, and disclosure requirements vary. Human authors remain accountable for accuracy, plagiarism prevention, citation, bias review, and the submitted work.[9]

Writing Uses Arranged by Risk

Use	Example	Risk level	Good practice
Brainstorming	Generate alternative section headings.	Lower	Retain only ideas that fit your argument.
Planning	Compare two outline structures.	Lower	Build the final structure from your research purpose.
Feedback	Identify unclear transitions or missing counterarguments.	Moderate	Decide whether each criticism is valid.
Language revision	Improve clarity in text you wrote.	Moderate	Compare before and after; protect technical meaning and voice.
Drafting prose	Produce a paragraph from your notes.	Higher	Check policy; rewrite, verify every claim, and disclose if required.
Generating citations or quotations	Create sources from model memory.	Unacceptable as evidence	Search and verify actual sources instead.
Writing with confidential material	Paste protected data or peer review into an unapproved tool.	High or prohibited	Use only an approved environment with permission and safeguards.

A Safe Writing Sequence

1. **Think first:** Write your claim, evidence, and purpose in your own notes.
2. **Ask for bounded assistance:** Request options, questions, critique, or clarity improvements.
3. **Compare:** Keep the original and the AI-assisted version.
4. **Verify:** Check meaning, evidence, citations, voice, originality, and policy.
5. **Revise yourself:** Produce the final text you can explain and defend.
6. **Document and disclose:** Record material use and follow course, institutional, funder, or publisher rules.

Prompt — Developmental Feedback, Not Ghostwriting

> *Read the paragraph below as a developmental editor. Do not rewrite it yet. Identify: (1) the apparent claim, (2) the evidence offered, (3) any unsupported leap, (4) ambiguous terms, (5) missing counterargument, and (6) one question the author should answer. Distinguish*

> *content problems from language problems. After I respond, propose two revision strategies while preserving my voice and citations.*

Prompt — Clarity Revision with Change Log

> *Improve the clarity and concision of my paragraph for **[audience]** without adding facts, citations, or claims. Preserve all technical meanings and citation markers. Return: (1) revised paragraph, (2) a change log explaining each substantive edit, and (3) any sentence whose meaning is uncertain. If the source text is already clear, say so rather than changing it unnecessarily.*

Activity 3.3 — Preserve Meaning

Original:

> *"The model's accuracy was higher after feature selection, although the small test sample and absence of external validation limit any conclusion about performance in other hospitals."*

Ask an AI tool to make the sentence "stronger and more persuasive." Then inspect the result.

Check	**Your observation**
Did it turn association into causation?	
Did it remove or weaken the limitations?	
Did it change "test sample" or "external validation" incorrectly?	
Is the revision more persuasive because it is clearer, or because it overclaims?	

Disclosure Template

Adapt this to the relevant policy; do not assume one statement fits every institution or publisher.

> *During preparation of this work, the author(s) used **[tool and version, if known]** on **[date or period]** for **[purpose, such as outline comparison, language revision, or code***

> ***explanation]***. *The tool was not used to* ***[important exclusions]***. *The author(s) reviewed and verified the output, checked all claims and citations against original sources, revised the material, and take full responsibility for the final work.*

3.4 AI-Assisted Coding Support

Coding assistants can generate boilerplate, explain code, suggest edits, propose tests, and help with documentation. Their output may still contain incorrect logic, weak tests, security defects, or unsupported dependencies, so human review and testing remain necessary.[6]

The Explain–Plan–Patch–Test–Review Loop

Stage	What to ask for	What you must do
Explain	Explain the code line by line, including inputs, outputs, and assumptions.	Confirm that the explanation matches the code.
Plan	Identify the likely cause and propose a minimal repair plan.	Decide whether the plan respects requirements.
Patch	Provide the smallest justified change; do not rewrite unrelated code.	Inspect every changed line and dependency.
Test	Generate normal, boundary, invalid-input, and regression tests.	Run tests in a controlled environment and add missing cases.
Review	Check security, performance, maintainability, license, and data handling.	Make the final acceptance decision and document it.

Activity 3.4 — Beginner Coding Lab

Consider this Python function:

```
def average_score(scores):
    total = sum(scores)
    return total / len(scores)
```

It works for [70, 80, 90], but what could go wrong?

Before using AI, predict at least two issues:

Now use this prompt:

> *Explain the Python function below to a beginner. State its intended input, output, and assumptions. Identify failures for an empty list, non-numeric values, and unexpected types. Do not change the code yet. Propose a clear input contract and four tests. After the explanation, give the smallest patch consistent with that contract. Do not silently discard invalid data.*

Record the result:

Review question	**Your judgment**
Did the explanation match the original code?	
Did the proposed contract make a defensible choice?	
Were normal, boundary, and invalid inputs tested?	
Did the patch introduce new dependencies or behaviour?	
Would you accept the code? Why or why not?	

Foundation Track — Plain-Language Algorithm

If you do not code, rewrite the function as instructions for a person. This exposes assumptions without requiring syntax knowledge.

Challenge Track — Requirements Matter

Compare these possible contracts:

- Reject an empty list with a clear error.
- Return None for an empty list.
- Return 0 for an empty list.

Which is correct? **None is universally correct.** The right behaviour depends on the application, downstream calculations, and documented requirements. An AI assistant cannot resolve an unstated requirement by confidence alone.

Secure Debugging Prompt

> ***I will provide a minimal, de-identified code example and the exact error. Do not request secrets, production data, access tokens, private keys, or unrelated files. First restate the expected behaviour and ask up to three clarification questions. Then identify likely causes ranked by evidence. Propose a minimal patch, explain each change, and create tests covering normal, boundary, invalid, and security-relevant inputs. Flag any library, API, or function name that you cannot verify.***

Coding Acceptance Checklist

Before using AI-generated code, confirm that you can answer **yes** to the relevant questions.

Question	Yes / No / N/A
I understand what every accepted line does.	
The code matches written requirements rather than guessed requirements.	
Inputs, outputs, errors, and side effects are defined.	
I ran normal, boundary, invalid-input, and regression tests.	

I checked security-sensitive behaviour and secret handling.	
Dependencies and API names exist and are appropriate.	
I checked performance where scale matters.	
I checked license, attribution, and organisational policy where relevant.	
No confidential data or credentials were exposed to an unapproved service.	
A qualified person reviewed consequential code before deployment.	

Reusable Prompt Library

1. Ask for Clarification before answering

> *My goal is **[goal]**, and the output will be used for **[decision or audience]**. Before answering, ask up to five questions needed to clarify scope, evidence, constraints, and risk. If a question cannot be answered from the provided material, identify it as missing rather than guessing.*

2. Evidence-Bounded question answering

> *Answer using only the supplied sources. For every substantive claim, provide the source name and page or section. Separate direct source statements from your inference. If the evidence conflicts, describe the conflict. If the answer is absent, write "insufficient evidence in the supplied sources."*

3. Research gap exploration

> *Based only on the verified study summaries in the table below, identify recurring limitations, under-studied populations, inconsistent measures, and unresolved findings. Do not call something a "research gap" merely because it appears in one paper. For each proposed gap, list the supporting studies and one alternative explanation.*

4. Reviewer Simulation

> *Review this draft as a constructive but critical peer reviewer for **[field or venue]**. Evaluate alignment among question, method, evidence, and conclusion. Separate major concerns, minor concerns, and clarification questions. Do not invent journal requirements. Quote the sentence or section that triggered each comment.*

5. Statistical Interpretation Check

> *Inspect the supplied results and proposed interpretation. Identify any confusion between statistical significance and practical importance, association and causation, prediction and explanation, subgroup and overall effects, or absence of evidence and evidence of absence. Do not calculate missing statistics. State what additional information is required.*

6. Writing Voice Preservation

> *Suggest clarity edits while preserving the author's terminology, caution level, and citation markers. Do not add claims. Mark every changed phrase and explain why it improves comprehension. Flag edits that may change meaning for author approval.*

7. Code Explanation

> *Explain this code at two levels: first for a non-programmer using an everyday analogy, then for a programmer using precise terminology. Identify inputs, outputs, state changes, dependencies, assumptions, time or memory concerns, and possible failure modes. Do not propose changes until the explanation is complete.*

8. Test Generation

> *Given the function specification and code, create a test table before writing test code. Include normal cases, boundary cases, invalid inputs, regression cases, and security-relevant cases. For each test, state input, expected result, and the requirement it checks. Then generate tests using **[framework]** without inventing unavailable dependencies.*

Responsible-Use Decision Card

Before you press **Send**, ask:

Question	Decision
Purpose: Is AI appropriate for this task, or do I need human expertise or an authoritative database?	
Permission: Do policy, consent, contract, copyright, and tool terms allow this use?	
Privacy: Is the information green, amber, or red?	

Prompt: Have I specified task, audience, sources, and key requirements?	
Proof: How will I test claims and review sources?	
People: Who may be excluded, misrepresented, or harmed?	
Provenance: Can I record the tool, date, prompt, sources, and important output?	
Ownership: Can I explain, defend, revise, and take responsibility for the final result?	
Disclosure: Does my institution, instructor, funder, employer, or publisher require acknowledgment?	

If you cannot answer a high-risk question, **pause rather than prompt**.

Personal Action Plan and Exit Ticket

Post-Session Self-Rating

Return to the diagnostic at the beginning and complete the **after** column. Which rating changed most, and why?

3–2–1 Exit Ticket

Three ideas I learned:

Two practices I will apply:

One question I still have:

My Seven-Day Experiment

Choose one low-risk, useful task you will test within the next seven days.

Item	**My plan**
Task	
Why AI may help	
Data classification	Green / Amber / Red
TASK prompt	
TRACE verification method	
Evidence of success	
Disclosure or permission needed	

> ***Commitment:*** *I will use AI to extend my judgment, not to avoid it.*

Model Debrief Notes

These are discussion guides, not the only acceptable answers.

Activity 1.1

Fraud prediction and adaptive scheduling are commonly ML applications; illustration generation is GenAI; chatbot summarisation and natural-language code generation are LLM applications and also GenAI. The important lesson is that one real system may fit several categories. Naming the model, application, data, and human decision separately improves accountability.

Activity 1.3

Brainstorming public, non-sensitive titles is usually lower risk. Exact quotations from an unseen paper require verification and should not be trusted from model memory. Identifiable patient data should not be placed in a public chatbot. Error explanations can be useful if the fix is understood and tested. Unopened AI-generated references are not evidence. Alternative paragraph structures are generally useful if meaning, voice, and citations are preserved.

Activity 2.1

A strong revision should narrow "social media," define the student population and outcome, specify the evidence sources and date range, request a structured synthesis, and state how unsupported claims and missing evidence must be handled. An AI-generated narrative is not a literature review unless the search, screening, appraisal, extraction, and synthesis process is defensible.

Activity 3.4

The original function fails on an empty list because division by zero is undefined in the code. Non-numeric elements can cause sum to fail, and a non-iterable input can also fail. However, the repair depends on the agreed contract. Returning zero may conceal missing data, while an explicit error or none may be more appropriate in some systems. Tests should reflect the intended behaviour rather than merely the generated implementation.

Glossary

The definitions below are practical workshop definitions, simplified from the cited foundational explanations and responsible-use guidance.[2] [3] [11]

Term	Working definition
Algorithm	A defined set of steps for solving a problem or performing a computation.
Artificial intelligence	The broad field of systems performing tasks such as prediction, recognition, generation, planning, or decision support.
Bias	A systematic tendency that can produce distorted, incomplete, or unfair outcomes.
Context window	The amount of input and generated material a model can consider during one interaction.
Deep learning	Machine learning based on multi-layer neural networks.
Embedding	A numerical representation that places related content near each other in a learned space.
Foundation model	A broad model adaptable to many downstream tasks.
Generative AI	AI designed to generate content such as text, images, audio, video, or code.
Hallucination / confabulation	Plausible but incorrect or unsupported generated content.
Inference	Using a trained model to generate a prediction or output for new input.
Large language model	A large model trained to process and generate language.
Machine learning	A method through which systems learn patterns from data.
Multimodal model	A model that processes or generates more than one type of content, such as text and images.
Prompt	Instructions and context given to a generative model.
Retrieval-augmented generation	A workflow that retrieves external information and supplies it to a model during generation. Retrieval can improve grounding but does not eliminate the need to verify.
Token	A text unit processed by a language model, often a word or part of a word.
Training data	Data used to adjust a model so it learns patterns relevant to its task.

References

Sources accessed: 14 July 2026. Official pages and tool features may change; verify current documentation before consequential use.

[1]. [Online]. Available: https://www.unesco.org/en/articles/guidance-generative-ai-education-and-research. [Accessed: Jul. 15, 2026].

[2]. National Institute of Standards and Technology, Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile, NIST AI 600-1, Gaithersburg, MD, USA, Jul. 2024, doi: 10.6028/NIST.AI.600-1.

[3]. Center for Security and Emerging Technology, "What are generative AI, large language models, and foundation models?," CSET. [Online]. Available: https://cset.georgetown.edu/article/what-are-generative-ai-large-language-models-and-foundation-models/. [Accessed: Jul. 15, 2026].

[4]. DAIR.AI, "General tips for designing prompts," Prompt Engineering Guide. [Online]. Available: https://www.promptingguide.ai/introduction/tips. [Accessed: Jul. 15, 2026].

[5]. OpenAI, "Prompt engineering," OpenAI API Documentation. [Online]. Available: https://platform.openai.com/docs/guides/prompt-engineering. [Accessed: Jul. 15, 2026].

[6]. GitHub, "Responsible use of GitHub Copilot code completion," GitHub Docs. [Online]. Available: https://docs.github.com/en/copilot/responsible-use/copilot-code-completion. [Accessed: Jul. 15, 2026].

[7]. Elicit, "AI for scientific research," Elicit. [Online]. Available: https://elicit.com/. [Accessed: Jul. 15, 2026].

[8]. ResearchRabbit, "Literature discovery and citation mapping," ResearchRabbit. [Online]. Available: https://www.researchrabbit.ai/. [Accessed: Jul. 15, 2026].

[9]. Princeton University Library, "Disclosing the use of AI," Research with Generative AI. [Online]. Available: https://libguides.princeton.edu/generativeAI/disclosure. [Accessed: Jul. 15, 2026].

[10]. University of Washington Graduate School, "Effective and responsible use of AI in research," UW Graduate School. [Online]. Available: https://grad.uw.edu/advice/effective-and-responsible-use-of-ai-in-research/. [Accessed: Jul. 15, 2026].

[11]. Center for Security and Emerging Technology, "The surprising power of next-word prediction: Large language models explained, part 1," CSET. [Online]. Available: https://cset.georgetown.edu/article/the-surprising-power-of-next-word-prediction-large-language-models-explained-part-1/. [Accessed: Jul. 15, 2026].

[12]. OpenAI, "Why language models hallucinate," OpenAI. [Online]. Available: https://openai.com/index/why-language-models-hallucinate/. [Accessed: Jul. 15, 2026].

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